Pairwise between-family d' calculations

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1 Data

1.1 Loading

See `20260625.qmd` for reference. For now, analysis will be limited to reliable sites and images with naturalistic structure.

2 LDA

2.1 Procedure

Lee et al. trained the population decoder by:

1. Taking a training set of repetitions (15-fold CV separated by sample) (**this notebook**).
2. Projecting the population response to non-blank images onto the first several principal components. Gerick mentioned this step was omitted in later analyses. Performing LDA on the original data works fine and removes the issue of finding PCs for combinations of subjects.
3. Applying linear discriminant analysis (LDA) to find the axis separating the naturalistic and noisy textures. For between-family discrimination, this is replaced by finding the axis best separating the families.
4. Projecting held-out responses on the axis, then reporting the d' on the resulting one-dimensional values.

2.2 Variations

Due to the low number of repetitions per individual image (6 repetitions), it is probably untenable to train LDA on pairwise sample comparisons. For low repetition analyses, I'll apply random dichotomies to increase the power.

LDA can be applied on:

1. Pairwise family comparisons with 15-fold CV (holdout is one of the sample IDs)
2. Random dichotomies within families with 6-fold CV (holdout is one of the repetition IDs)

3. Random dichotomies between families with 6-fold CV (similar to the previous variation but with samples taken from different families)

2.3 Procedure for this notebook

I subsampled 30 sites for each region out of the reliable sites. Subsampling did not consider simultaneous recording, so the subsampled sites may include more than one array. I fit LDA with `MASS::lda()` and found the projections with the `x` output of `predict.lda()` on the test set.

3 Pairwise between-family distances

For each family pair, data is saved as a list with:

- `dprimes`: a data frame
  - `mu`: the difference in means
  - `sd`: the combined standard deviation
  - `data_split`: the `trn` or `tst` reading
  - `dprime`: the calculated Cohon's d', found as `mu` / `sd`
  - `k`: the label for the fold in 15-fold CV (corresponding to 15 samples)
- `scalings`: a numeric 15×73 matrix corresponding to 15 samples by the number of sites (here, determined by the number of reliable sites)
- `group_ids`: a numeric vector containing the group IDs included in the pair

Within this notebook, I mostly sanity tested some of the procedures. See `scripts/pairwise_family.R` for the code. I ran the calculations in the Stats server.

d' values are reported as the average across the 15 folds. See Figure 1 for histograms of the pairwise comparison d' values (subsampled for 30 sites), separated by region. The trend in the results seem consistent with Gerick's poster (copied in Figure 2 for reference), with the center of the histogram generally tending toward higher d' with higher visual regions with the trend reversing in IT. Both the difference in means and combined SD tend to increase with higher visual regions, again with the trend reversing in IT. This may indicate that for V1 to V2 to V4, the firing patterns tend to be more spread out, with the increased variance compensated by better separation of the groups. There does not seem to be a notable difference in train and test d' (Figure 1, right).

Though the relative differences in d' distributions seem similar to Gerick's previous results, the scales are inconsistent, with Gerick finding d' values as high as around 6 for V4 (my max: 1.3). Possibly, he calculated LDA on averages over repetitions, thus lowering the combined SD (the denominator of the d'). It may also be due to differences in the number of sampled sites.

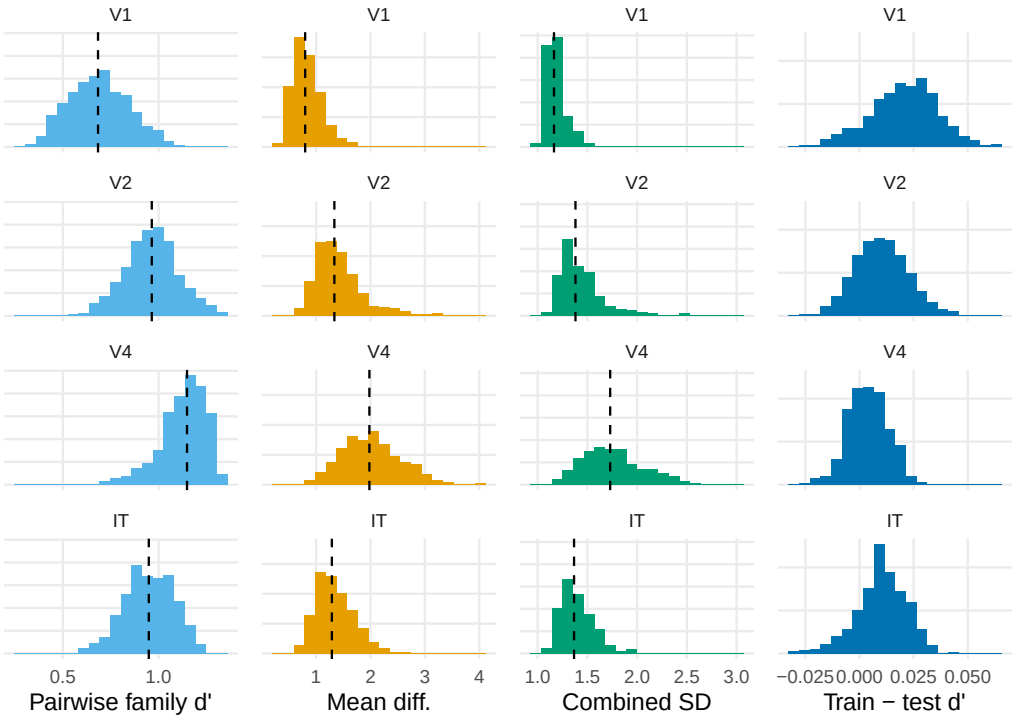


Figure 1: Histogram of d', mean differences, combined SD, and train-test difference for each pairwise family comparison, separated by region. Vertical line indicates the median.

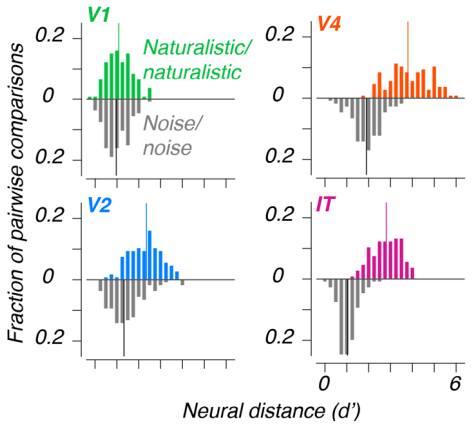


Figure 2: Histograms of pairwise comparisons, for naturalistic-naturalistic comparisons (colored bars), and noise-noise comparisons (gray). Lines depict medians. The figure was originally vertical but I rearranged it, so the axes are consistent for all quadrants.